# Analysis of Risk Factors for Type II Diabetes: A Robust and Multivariate Approach


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**ABSTRACT:**

This project investigates the main factors associated with Type II diabetes using a dataset of 403 individuals from a medical study conducted in Virginia. The analysis focuses on the role of obesity-related measures and cardiovascular risk factors in predicting diabetes. Given the clinical nature of the data, the dataset presents challenges such as noise, non-clear outliers, and complex variable relationships, making the modelling task complex.

A baseline logistic regression model is first estimated and then refined through variable selection and transformations. To address data irregularities, robust methods are applied. In addition, different modelling approaches, including linear models and random forest, are considered to compare alternative predictive frameworks. Clustering methods, including k-means, trimmed k-means and model-based clustering, are also implemented to identify groups of individuals with similar risk profiles.

Model performance is evaluated using confusion matrices, ROC curves, AUC and accuracy. The results confirm the importance of obesity-related variables and show that combining robust and flexible methods allows us to obtain reliable and interpretable results, despite the complexity of the dataset.

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# 1. Introduction

Type II diabetes is one of the most common health issues and is often associated with obesity and cardiovascular risk factors. The relationships among these variables are difficult to understand, particularly when working with real data: typically affected by noise, outliers and non-linear patterns.

The goal of this project is to identify the main factors associated with diabetes using a multivariate statistical approach, with special attention given to robust methods. The analysis will not be focused on predictive performance, as it will be on the comparison of different modeling techniques and their ability to handle data irregularities.

Therefore, no train-test split or cross-validation procedures are applied, as we are not interested in maximizing predictive accuracy, but in highlighting the behavior and robustness of different statistical methods in a controlled setting.

# 2. Data Description

## 2.1 Dataset Overview

The dataset used in this analysis consists of 403 individuals screened for Type II diabetes as part of a medical study conducted in central Virginia. The original study aimed to investigate the prevalence of obesity, diabetes and other cardiovascular risk factors within a specific population.

It is fundamental to highlight that the data are derived from a real clinical context. As a result, the dataset presents several challenges, including potential measurement variability, the presence of outliers and complex relationships among variables. These aspects make the analysis more realistic but also require careful modeling choices and robust statistical techniques.

## 2.2 Variables Description

The dataset includes 19 variables describing demographic characteristics, anthropometric measures and clinical indicators related to diabetes and cardiovascular risk.

Demographic variables include age, gender (male/female), and location, that identifies the area where the subject was located in the study; the variable frame classifies individuals according to body frame size (small, medium, large).

Anthropometric variables describe body composition and include height (in inches), weight (in pounds), waist and hip circumference (in inches). These measures are particularly relevant for the analysis, as they enable the derivation of indicators associated with obesity and fat distribution, which are strongly correlated with Type II diabetes.

The clinical variables capture metabolic and cardiovascular conditions: these include total cholesterol (chol), high-density lipoprotein (hdl), and their ratio, along with stabilized glucose (stab.glu) and glycosylated hemoglobin (glyhb), which is used to determine diabetes status. Blood pressure is measured through systolic and diastolic values (bp.1s, bp.1d, bp.2s, bp.2d), providing multiple observations for each individual; the variable time.ppn represents the time passed after food intake when laboratory measurements were taken.

There are also some data quality issues in this dataset, which are common in clinical studies: some variables contain missing values, with the second blood pressure measurements (bp.2s and bp.2d) showing a high proportion of missing observations. Other variables exhibit moderate levels of missingness or potential measurement variability. These aspects are all taken into account in the following analysis

**Image 1: initial dataset**

| Name | Label | Units | Levels | Type | NAs |
|---|---|---|---|---|---|
| **id** | Subject ID | | | Continuous | 0 |
| **glyhb** | Glycosylated Hemoglobin | | | Continuous | 13 |
| **stab.glu** | Stabilized Glucose | | | Continuous | 0 |
| **chol** | Total Cholesterol | | | Continuous | 1 |
| **hdl** | High Density Lipoprotein | | | Continuous | 1 |
| **ratio** | Cholesterol/HDL Ratio | | | Continuous | 1 |
| **age** | Age | years | | Continuous | 0 |
| **time.ppn** | Postprandial Time (labs drawn) | minutes | | Continuous | 3 |
| **height** | Height | inches | | Continuous | 5 |
| **weight** | Weight | pounds | | Continuous | 1 |
| **bp.1s** | First Systolic Blood Pressure | | | Continuous | 5 |
| **bp.1d** | First Diastolic Blood Pressure | | | Continuous | 5 |
| **waist** | Waist Circumference | inches | | Continuous | 2 |
| **hip** | Hip Circumference | inches | | Continuous | 2 |
| **bp.2d** | Second Diastolic Blood Pressure | | | Continuous | 262 |
| **bp.2s** | Second Systolic Blood Pressure | | | Continuous | 262 |
| **gender** | Gender | | 2 | Categorical | 0 |
| **frame** | Body Frame Size | | 3 | Categorical | 12 |
| **location** | Study Location | | 2 | Categorical | 0 |

*For clarity: the first systolic and diastolic blood pressure measurements (bp.1s and bp.1d) were later renamed systolic and diastolic, and used in place of the original notation in the subsequent analysis.*
*The second measuring variables (bp.2d, bp.2s) will be removed for too many N/A*

# 3. Exploratory Data Analysis

## 3.1 Initial Data Exploration

The dataset was preprocessed in addressing missing values, define the response variable, and construct more informative predictors.

Firstly, the target variable was defined according to glycosylated hemoglobin. Based on the clinical convention mentioned in the description of the dataset, individuals with glyhb > 7 were considered diabetic, while all others were considered non-diabetic. This led to the creation of a binary response variable used in all subsequent supervised analyses (84% of total: non-diabetic).

Regarding missing values, the second blood pressure measurements (bp.2s and bp.2d) were removed due to the very high percentage of missing values (over 60% of the sample).
It's also important to say that their inclusion wouldn't really provide additional information beyond the first blood pressure measurements. After that, observations with the remaining missing values (very limited, see Image 1) were removed, leading to a complete dataset.
At this point, an initial logistic regression model was estimated using all the available variables to assess their joint contribution. The results showed that only stabilized glucose (stab.glu) appeared to be statistically significant. As expected, this suggested the presence of strong multicollinearity among the explanatory variables, which was further confirmed through the Correlation and VIF analysis (Images 2 and 3)

A number of variables have been transformed to improve interpretability and reduce redundancy among predictors: Body Mass Index (**BMI**) was computed from height and weight, Waist-to-Hip Ratio (**WHR)** from waist and hip circumference: a more direct representation of obesity and fat distribution compared to the original variables, that from now on will be excluded. Different the reasoning for **Ratio,** which is calculated directly as "Chol"/"Hdl": in many point of the study the approach to these three variables changed, trying to understand in which way the performance was better: most of the time we agreed to delete ratio and keeping the other two: to save more information while avoiding multicollinearity. A binary indicator of **hypertension** was introduced based on the thresholds for systolic and diastolic blood pressure (calculated as well as a continuous variable only for the Correlation matrix in image 4).

As a result, two versions of the dataset were included in the analysis: both developed after the removal of incomplete features and missing observations. The first one, referred to as the *Original Dataset*, includes the original variables. The second one: *Transformed Dataset*, replaces groups of correlated variables with their indicators ( BMI, WHR and hypertension) excluding the original variables used in their construction. This distinction will be maintained in the following sections to compare model performance under different specifications.

## 3.2 Correlation Analysis

In order to further examine the relationships among variables, a correlation analysis was conducted on *Original Dataset* using a correlation matrix (Image 2).
The results showed the presence of strong correlations among several predictors: variables related to body composition, such as weight, waist and hip circumferences, show very high positive correlations, indicating that they capture similar information.
A similar pattern is observed among lipid-related variables, where HDL and the cholesterol ratio show a strong inverse relationship, while systolic and diastolic blood pressure are moderately correlated, suggesting that they partially overlap.
These findings confirmed the presence of multicollinearity in the dataset, which was also evident from the initial regression model and the VIF analysis.

*Original Dataset*

**Image 2: Correlation matrix**

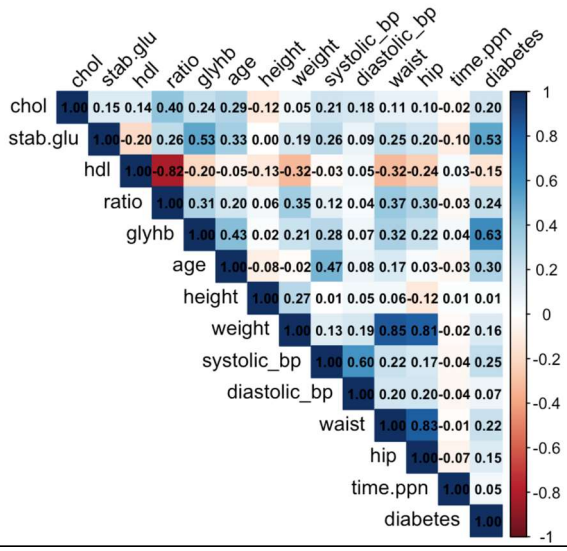


**Image 3: VIF**

| | GVIF | Df | GVIF^(1/(2*Df)) |
|---|---|---|---|
| chol | 3.160977 | 1 | 1.777914 |
| stab.glu | 1.258006 | 1 | 1.121609 |
| hdl | 5.803518 | 1 | 2.409049 |
| ratio | 6.158580 | 1 | 2.481649 |
| location | 1.304526 | 1 | 1.142159 |
| age | 1.852217 | 1 | 1.360962 |
| gender | 2.808500 | 1 | 1.675858 |
| height | 2.624021 | 1 | 1.619883 |
| weight | 6.845962 | 1 | 2.616479 |
| frame | 2.171518 | 3 | 1.137961 |
| systolic_bp | 2.350040 | 1 | 1.532984 |
| diastolic_bp | 2.083855 | 1 | 1.443556 |
| waist | 5.196914 | 1 | 2.279674 |
| hip | 5.465208 | 1 | 2.337778 |
| time.ppn | 1.107107 | 1 | 1.052191 |

*Transformed Dataset*

**Image 4: Correlation matrix**

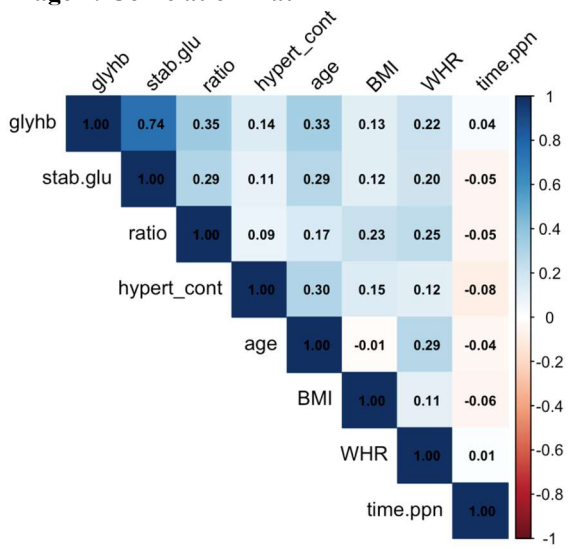


**Image 5: VIF**

| | GVIF | Df | GVIF^(1/(2*Df)) |
|---|---|---|---|
| stab.glu | 1.114108 | 1 | 1.055513 |
| ratio | 1.139159 | 1 | 1.067314 |
| location | 1.250404 | 1 | 1.118215 |
| age | 1.264401 | 1 | 1.124456 |
| gender | 1.540452 | 1 | 1.241150 |
| BMI | 1.516352 | 1 | 1.231402 |
| frame | 1.904343 | 3 | 1.113331 |
| WHR | 1.322376 | 1 | 1.149946 |
| hypert_cont | 1.149995 | 1 | 1.072378 |
| time.ppn | 1.074896 | 1 | 1.036772 |

A similar analysis was performed on the *Transformed Dataset* (Image 4): in this case Ratio was keeped while “hdl“ and “chol “removed, but all of the three are still present in the

dataset. This was only another way to study the influence of ratio in the dataset.
Compared to the original specification, the correlation structure appears significantly simplified. The strong dependencies previously observed among anthropometric variables are no longer present, as they are replaced by BMI and WHR. Overall, correlations among predictors are more moderate, indicating a reduction in redundancy.
This improvement is further confirmed by the VIF analysis (Image 5), where all variables exhibit low values, suggesting that multicollinearity has been substantially reduced.

## 3.3 Initial Insights

The transformed dataset shows a more compact representation of body composition, reducing redundancy while preserving relevant information.

Graphical exploration further highlights the role of key predictors: “stab.glu” and, obviously, “glyhb” (≈ target variable) show their strong explanatory power separeting well the two classes. In contrast, most other variables do not exhibit a clear separation suggesting that the classification problem is driven by a limited number of highly informative predictors.

The absence of well-defined clusters in the pairwise plots suggests that the data don’t naturally partition into distinct groups. This indicates that clustering methods may capture only weak underlying structures, rather than clear segmentations of the population.

Another thing that can be seen is that the presence of extreme observations across several variables confirms that the dataset contains, potentially, many outliers. This is particularly relevant in the context of clinical data and motivates the use of robust estimation techniques.

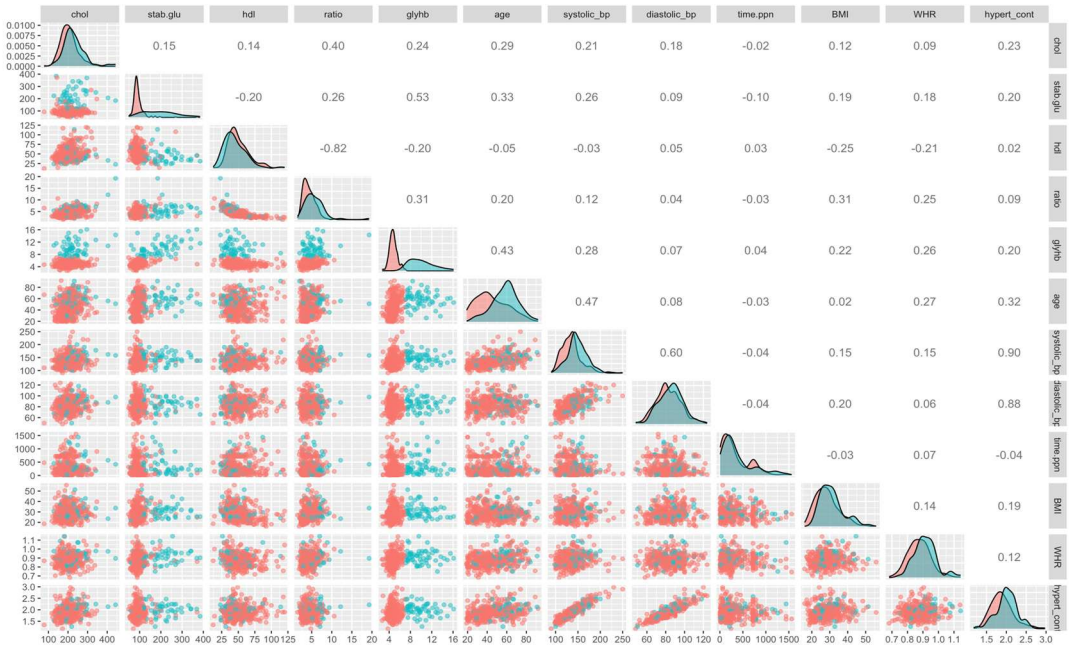


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**Image 6: visual 2D distributions**

# 4. Logistic Regression Analysis

## 4.1 Initial Specifications

The logistic regression analysis begins with an initial specification that includes all the available variables after the preliminary data preparation phase.

The first model (model1), already discussed, was estimated using the *Original Dataset* with the aim of evaluating their joint contribution to the probability of diabetes. As previously anticipated, the results showed that only stabilized glucose (stab.glu) appeared to be statistically significant.

Based on the findings discussed in 3.1 and 3.2, a second model specification was introduced. This refined model incorporates the transformed variables discussed: "BMI", "WHR", "Ratio" and "Hypertension", and excludes redundant predictors (this is the *Transformed Dataset*). This model shows a better performance, with more significant variables contributing to the explanation of the outcome. In particular, "stabilized glucose" remains strongly deterministic, while "age" also emerges as a relevant predictor. Intercept is now significative too.

Overall, the analysis started giving results, but the number of outliers is high and accuracy is still low: this is why an analysis of outliers was conducted using robust statistical techniques.

## 4.2 Outlier Detection using MCD

A slight modification of the *Transformed Dataset* was introduced: the variable "Ratio" was removed and replaced by its components "chol" and "hdl", this choice allows us to retain more information on lipid profiles while maintaining an acceptable level of multicollinearity.

The Minimum Covariance Determinant (MCD) estimator was applied to the *Transformed dataset*[1]. This approach provides robust estimates of the multivariate center by focusing on the most central subset of observations reducing the influence of extreme values, that is particularly important in a clinical context where extreme values are common.

Using the robust covariance estimates, Mahalanobis distances were computed for each observation, then a theoretical threshold based on the chi-squared distribution was used to identify outliers. The analysis reveals that a big number of observations exceed this threshold,

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[1] Since MCD requires continuous variables, we didn't consider the categoricals variables: except for "Hypertension" that was replaced by the original continuous measures "systolic_bp" and "diastolic_bp".

with 70 out of 375 observations classified as outliers (image 8). This confirms that extreme values are not isoletad, but a structural feature of the dataset.

Graphical inspection further supports these finding: the distance-distance plot (img 7) highlights clear deviations between classical and robust distances. These patterns indicate the presence of influential points that significantly affect standard estimation procedures.

**Image 7: Distance-Distance Plot**

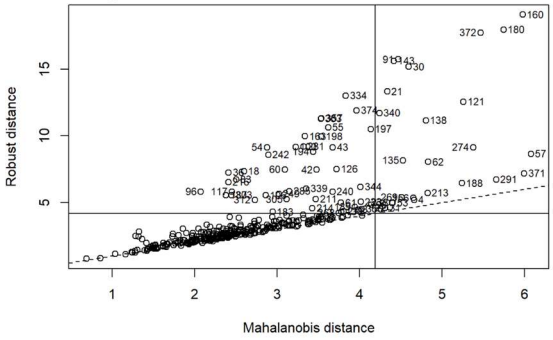
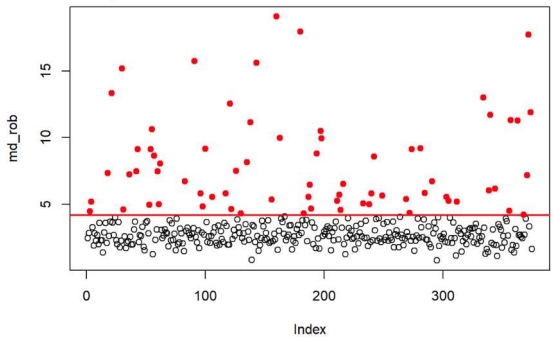



**Image 8: Outliers detection**




## 4.2.1 MCD Logistic in *Transformed Dataset*

The observations identified as outliers with MCD were removed from *Transformed Dataset.* This resulted in a reduction of the sample size from 375 to 305 observations.

A logistic regression model was then estimated on the *Transformed Cleaned Dataset*[2] . The results confirm the strong predictive role of stabilized glucose and age. However, the other variables are not significant.

Although coefficient estimates appear slightly more stable, the model still struggles to identify clear relationships beyond the main predictors. This indicates that the presence of outliers is not the only factor affecting model performance.

## 4.2.2 MCD Logistic in Clean Dataset

Another try was given at MCD method, focusing only on the most relevant variables. Categorical variables: “gender”, “location” and “frame” were excluded, as they appeared totally not relevant in previous models and are not strongly supported from a clinical perspective.

The results improve the previous performance a little, with also Intercept being significative and Cholesterol close to be, and all other variables not statistically significant.

The simplified specification improves the interpretability of the model and multicollinearity is no longer a concern, as confirmed by low VIF values. However, the overall performance

[2] in the code this model is called “complete_logit_clean”

remains unsatisfactory. For this reason, a different approach is required.
In the next section, robust logistic regression is introduced in order to account for the presence of influential observations without discarding them.

## 4.3 Robust Logistic Regression

Robust logistic regression allows us to reduce the influence of extreme and potentially influential observations without removing them from the dataset.

For this reason, robust models were estimated both on the Transformed Dataset, including all observations, and on a simplified specification, in order to evaluate whether robustness can improve model stability and interpretability.

### 4.3.1 Robust Transformed Dataset

A robust logistic regression model was first estimated on the Transformed Dataset, including all observations (in the code: *robust_logit_complete)*. In this specification, both the variable "ratio" and its components "chol" and "hdl" were retained. While this introduces some redundancy, it allows the model to capture different aspects of the lipid profile. In a robust framework, this is less problematic.

The results show a more stable structure compared to the MCD approach in logistic regression. Stabilized glucose remains the main predictor, while other variables exhibit more consistent estimates (img 9).

**Image 9**

```
## Coefficients:
##                  Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -19.714501   7.028820  -2.805  0.00503 **
## chol             0.053179   0.018865   2.819  0.00482 **
## stab.glu         0.090901   0.021046   4.319 1.57e-05 ***
## hdl             -0.124603   0.060705  -2.053  0.04011 *
## ratio           -1.280907   0.532605  -2.405  0.01617 *
## locationLouisa   0.740701   0.900175   0.823  0.41060
## age              0.012014   0.025259   0.476  0.63434
## gendermale      -2.051412   1.167684  -1.757  0.07895 .
## BMI              0.055451   0.054873   1.011  0.31224
## framelarge      -0.822811   2.331715  -0.353  0.72418
## framemedium     -0.856287   2.367381  -0.362  0.71757
## framesmall      -0.660827   2.604126  -0.254  0.79968
## WHR              5.061234   5.458161   0.927  0.35378
## hypertension1    1.714752   0.897656   1.910  0.05610 .
## time.ppn         0.002330   0.001072   2.174  0.02968 *
```

**Image 10**

```
## Coefficients:
##                  Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -18.676250   5.280037  -3.537 0.000404 ***
## BMI             0.083950   0.038500   2.181 0.029219 *
## WHR            -0.730662   4.024488  -0.182 0.855933
## stab.glu        0.060967   0.011540   5.283 1.27e-07 ***
## chol            0.025506   0.009607   2.655 0.007932 **
## hdl            -0.004920   0.016643  -0.296 0.767513
## hypertension1   1.044872   0.702069   1.488 0.136678
## age             0.020278   0.018416   1.101 0.270866
## time.ppn        0.001781   0.000867   2.054 0.039990 *
```

### 4.3.2 Robust Clean Dataset

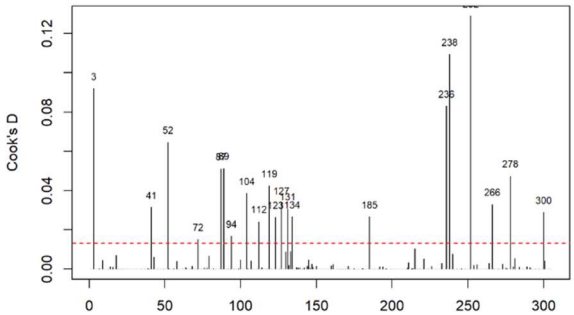
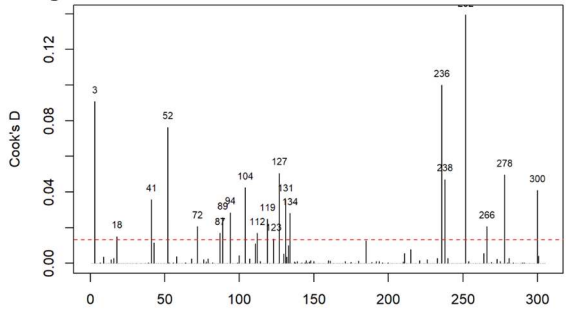
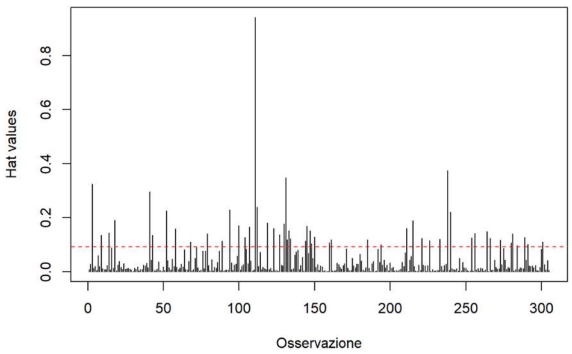
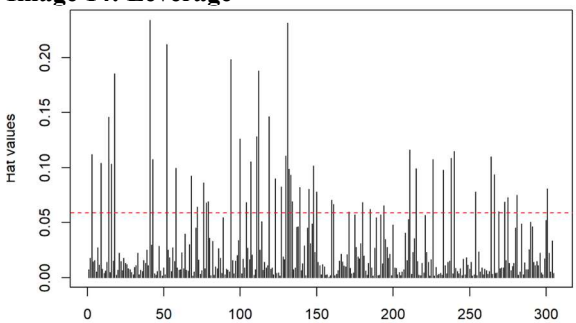
A second robust logistic (in the code: *robust_logit*) regression model was estimated focusing only on the most relevant variables (as in 4.2.2): the results show a further improvement in terms of stability and interpretability. Coefficient estimates are mostly significative and easier to interpret (image 10).

The robust approach limits the influence of extreme observations without requiring their removal, generating so far, the best models.

## 4.4 Final Model Comparison

In this section, the performance of the main models is compared. The initial specifications are not included in the comparison, as they were mainly exploratory: the analysis focuses on the models estimated after the MCD procedure (clean dataset) and on the robust logistic models. The comparison is conducted on both diagnostic plots and performance metrics.

### 4.4.1 Model diagnostics (plots)

| MCD model 1 (4.2.1) | MCD model 2 (4.2.2) |
|---|---|
| **Image 11: Cook's distance**   | **Image 13: Cook's distance**   |
| **Image 12: Leverage**   | **Image 14: Leverage**   |

To further assess model stability, diagnostic measures based on Cook's distance and leverage were analyzed: useful to identify points that may have a disproportionate impact on the estimated coefficients. The plots (Images 11 and 13) show the presence of influential points in both models. Similarly, the leverage plots (Images 12, 14) highlight observations with high values, corresponding to individuals with atypical combinations of predictors.

Cook's distance shows similar pattern in the two models, while the leverage structure differs more substantially. In the simplified specification (4.2.2), leverage values are generally higher and more dispersed: this can be explained by the reduction in the number of variables: by focusing only on the most relevant predictors, the model operates in a lower-dimensional space, making some observations appear more extreme in terms of their predictor values.

### 4.4.2 Performance metrics (accuracy, AUC, etc.)

| | Model | Accuracy | Precision | Recall | Specificity | F1-score | AUC |
|---|---|---|---|---|---|---|---|
| MCD | complete_logit_clean (4.2.1) | 0.944 | 0.667 | 0.111 | 0.997 | 0.191 | 0.875 |
| | logit_clean (4.2.2) | 0.944 | 0.667 | 0.111 | 0.997 | 0.191 | 0.857 |
| ROBUST | robust_logit_complete (4.3.1) | 0.936 | 0.840 | 0.724 | 0.975 | 0.778 | 0.920 |
| | robust_logit (4.3.2) | 0.939 | 0.857 | 0.724 | 0.978 | 0.785 | 0.928 |

**Image 15**

The results reported in *Image 15* show clear differences across model specifications.

The models estimated after the MCD procedure achieve high accuracy but show very low recall and F-1 score. While precision is relatively high, these models correctly identify only a small fraction of diabetic individuals. This indicates that they are highly conservative in predicting the positive class and fail to properly distinguish between the two classes.

In contrast, robust logistic models show a substantial improvement. Recall increases significantly, meaning that a much larger proportion of diabetic individuals is correctly identified, while precision remains high. As a result, the F1-score improves considerably, indicating a better balance between false positives and false negatives.

Between the two robust specifications, the simplified model (robust_logit) slightly outperforms the complete one. It achieves higher precision, F1-score and AUC, while maintaining similar recall. This suggests that removing less relevant variables improves model stability and predictive performance.

Even remembering that no train-test split or cross-validation was applied (see introduction) and the reported performance may therefore be optimistic, the results are still better than expected given the complexity of the original dataset.

# 5. Other specifications

## 5.1 Linear Models and Initial Specifications

Since the logistic regressions provided somehow interesting but not overly satisfactory results, we decided to remodel the previously computed variable diabetes to obtain again the variable glyhb, in order then to compute different robust and non-robust linear regressions following the same pipeline used before. The idea behind this reasoning was simple: treating Hemoglobin as a dummy may lead to a huge loss of information, especially for the observations that fall slightly below or above the threshold, making then results possibly less reliable. With a continuous variable instead, we can preserve the full variability of data, hopefully obtaining models with increased statistical power and improved performance. Moreover, with this approach we give a more realistic representation of the underlying biological process that leads to the diagnosis, that is inherently continuous rather than discrete. On the other hand, we should consider that if we apply this more flexible framework, we are trading off some of the interpretability of the results of the model, since we cannot immediately interpret coefficients in terms of their contribution to the risk of being positively diagnosed with diabetes or not.

As anticipated, the pipeline followed is very similar to the one applied in the logistic model section. Firstly, two very simple and general models have been computed: the first one (lm1) included all the variables of the dataset before any kind of transformation of already existing variables, the second one (lm2) instead included the newly generated ones, that are BMI, WHR and hypertension. This second specification was motivated again by the fact that composite indicators may better capture underlying physiological mechanisms compared to their individual components. Of course, these two very simple models are not expected to produce satisfactory and meaningful results due to possible problems of multicollinearity, but they serve as a baseline and a starting point to improve then performance and predictive capabilities of our subsequent models.

The two models' specifications and their results are listed below:

```
Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)   -2.5408106  2.2879923  -1.110   0.2675
chol           0.0041589  0.0032587   1.276   0.2027
stab.glu       0.0274573  0.0015006  18.298   <2e-16 ***
hdl           -0.0009551  0.0100674  -0.095   0.9245
ratio          0.1074493  0.1115257   0.963   0.3360
locationLouisa -0.2241167  0.1575545  -1.422   0.1558
age            0.0129350  0.0059161   2.186   0.0294 *
gendermale    -0.1079460  0.2449762  -0.441   0.6597
height         0.0208417  0.0296639   0.703   0.4828
weight        -0.0040443  0.0051957  -0.778   0.4369
framelarge     0.3861209  0.5131395   0.752   0.4523
framemedium    0.6544899  0.5020623   1.304   0.1932
framesmall     0.5935538  0.5175051   1.147   0.2522
systolic_bp    0.0026343  0.0047750   0.552   0.5815
diastolic_bp  -0.0015519  0.0074402  -0.209   0.8349
waist          0.0312572  0.0302725   1.033   0.3025
hip            0.0144394  0.0344868   0.419   0.6757
time.ppn       0.0005639  0.0002465   2.287   0.0228 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

```
Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)   -0.9755181  1.2405218  -0.786  0.43216
chol           0.0042807  0.0032327   1.324  0.18628
stab.glu       0.0276209  0.0014870  18.575  < 2e-16 ***
hdl           -0.0008226  0.0100218  -0.082  0.93463
ratio          0.1090305  0.1109464   0.983  0.32640
locationLouisa -0.2316046  0.1546685  -1.497  0.13516
age            0.0161147  0.0052077   3.094  0.00213 **
gendermale    -0.0985372  0.1788581  -0.551  0.58203
BMI            0.0102082  0.0141485   0.722  0.47107
framelarge     0.3292410  0.5084350   0.648  0.51768
framemedium    0.6012725  0.4960215   1.212  0.22623
framesmall     0.5214768  0.5105433   1.021  0.30774
WHR            0.8805306  1.1804841   0.746  0.45621
hypertension  -0.1380059  0.1620169  -0.852  0.39489
time.ppn       0.0005245  0.0002448   2.143  0.03282 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

As expected, results do not seem to be very exciting: both the models provided only three statistically significant estimates, out of the 18 and 15 computed. Therefore, at least for this initial analysis, only the level of glucose, the age and the post prandial time when labs were

drawn seem to have a substantial effect. Moreover, the significant coefficients were found to be very close to zero, suggesting an impact on the diabetes status, but of very low magnitude.

To check if the outcomes could improve, we decided to run another simple linear regression model (lm3) on a subset of variables, that excluded the categorical ones that seemed not to have an impact on our phenomenon, and the variable ratio, that was responsible of causing problems of multicollinearity. The result is the following:

```
Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)  -0.1135469  1.0319207  -0.110 0.912442
BMI           0.0076398  0.0118557   0.644 0.519720
WHR           0.7953038  1.0704134   0.743 0.457966
stab.glu      0.0276827  0.0014729  18.795  < 2e-16 ***
hypertension -0.1391530  0.1609159  -0.865 0.387739
chol          0.0070081  0.0017839   3.928 0.000102 ***
hdl          -0.0088612  0.0046000  -1.926 0.054837 .
age           0.0146950  0.0050726   2.897 0.003995 **
time.ppn      0.0005730  0.0002401   2.387 0.017490 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Fortunately, the results improved significantly. In this last specification, 4 out of the 9 coefficients computed achieved a statistically relevant meaning, with the coefficient of hdl that is very close to the significance level of 0,05. In particular, some variables remain the same as before, as for example time.ppn, age and stab.glu, but now we have also a significant effect of cholesterol. Although the outcome in terms of statistical significance improved, we still get coefficients very close to zero, limiting the impact of a unitary variation of these variables on the level of Glycosolated Hemoglobin of an individual.

For these reasons, we tried to model other specifications, in order to possibly get better results.

## 5.2 Linear Models after Outlier Detection with MCD

The first approach followed was again of detecting and deleting from our dataset all the outliers, running then similar models to the ones already discussed to check whether using the central structure of our data increases efficiency and performance.

For outlier detection, the results of the previously applied Minimum Covariance Determinant (MCD) were used. For this reason, the number of outliers is still 70, and the clean dataset used for running the regressions is the same as before.

The two models computed use the same variables used in the logistic case: the first (lm4) contains all the variables obtained after the transformations:

```
Coefficients:
                 Estimate Std. Error t value Pr(>|t|)
(Intercept)    -0.0718164  1.3875652  -0.052 0.958758
chol            0.0003691  0.0042363   0.087 0.930637
stab.glu        0.0173847  0.0045380   3.831 0.000157 ***
hdl             0.0059743  0.0144366   0.414 0.679304
ratio           0.1121808  0.1671036   0.671 0.502548
locationLouisa -0.1552796  0.1442675  -1.076 0.282674
age             0.0237070  0.0051228   4.628 5.58e-06 ***
gendermale      0.0118520  0.1673848   0.071 0.943600
BMI             0.0179918  0.0137483   1.309 0.191688
framelarge      0.3885642  0.5163748   0.752 0.452370
framemedium     0.5616067  0.5074255   1.107 0.269308
framesmall      0.4511932  0.5217749   0.865 0.387903
WHR             0.8490904  1.1807022   0.719 0.472634
hypertension   -0.2163316  0.1480798  -1.461 0.145122
time.ppn        0.0001453  0.0002349   0.618 0.536776
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

While the second (lm5) uses the same subset of variables used for lm3:

```
Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)   0.6435835  1.1098940   0.580 0.562450
BMI           0.0155268  0.0115689   1.342 0.180588
WHR           1.0121932  1.0475640   0.966 0.334715
stab.glu      0.0171262  0.0044982   3.807 0.000171 ***
chol          0.0031747  0.0019017   1.669 0.096099 .
hdl          -0.0023764  0.0052381  -0.454 0.650398
hypertension -0.2080354  0.1454661  -1.430 0.153734
age           0.0226214  0.0049505   4.570 7.18e-06 ***
time.ppn      0.0001964  0.0002268   0.866 0.387128
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Interestingly, the models estimated on the cleaned dataset exhibited slightly worse performance compared to those fitted on the original data. We can in fact notice that the number of coefficients found to be significant is lower, even if their statistical relevance is higher. While it may appear counterintuitive, these results can come from different aspects. First, the observations identified as outliers may still contain meaningful information rather than being purely erroneous, and their “extreme” values may still be very useful, especially in medical contexts as ours.

Second, the exclusion of outliers reduces both the variability and the size of the dataset. This can limit the model’s ability to capture the full range of relationships between predictors and the outcome, ultimately leading to lower predictive performance.

Overall, the outcomes obtained by this second analysis were not completely satisfactory. For this reason, we proceeded with another robust approach.

## 5.3 Robust Linear Models

To try to improve our results we decided then to rely on two robust linear models, using the same specifications of the last-mentioned ones. The reason is that unlike the MCD-based approach, which removes extreme observations, the robust regression model retains all data points while reducing the influence of outliers through a weighting mechanism, and this peculiarity seemed to fit very well the needs of our analysis.

In doing such a thing, we decided to rely on the r package MASS, and we stuck with the default settings that fit the model using M-estimation with a Huber loss function. This loss

function was preferred over more aggressive alternatives, such as Tukey's function, because it provides a more conservative approach to handling outliers. In this framework, extreme observations are down weighted but not completely discarded, allowing the model to retain potentially meaningful information. We thought that this approach could be better, especially considered the health-related context of our study, where extreme values can often be of interest. Then the estimation of the parameter is conducted with the classical iterative procedure until convergence.

The specification of the first model follows the same one of lm2 and lm4, while the second one is the same of lm3 and lm5. In the end, the results obtained are the following:

Coefficients:

| | Value | Std. Error | t value |
|---|---|---|---|
| (Intercept) | -0.5672 | 0.8381 | -0.6768 |
| chol | 0.0029 | 0.0022 | 1.3072 |
| stab.glu | 0.0295 | 0.0010 | 29.4108 |
| hdl | 0.0003 | 0.0068 | 0.0483 |
| ratio | 0.0786 | 0.0750 | 1.0483 |
| locationLouisa | -0.2546 | 0.1045 | -2.4369 |
| age | 0.0126 | 0.0035 | 3.5757 |
| gendermale | -0.0731 | 0.1208 | -0.6048 |
| BMI | 0.0119 | 0.0096 | 1.2449 |
| framelarge | 0.2642 | 0.3435 | 0.7692 |
| framemedium | 0.3789 | 0.3351 | 1.1307 |
| framesmall | 0.3466 | 0.3449 | 1.0048 |
| WHR | 0.8525 | 0.7975 | 1.0689 |
| hypertension | -0.1402 | 0.1095 | -1.2808 |
| time.ppn | 0.0004 | 0.0002 | 2.4045 |

Coefficients:

| | Value | Std. Error | t value |
|---|---|---|---|
| (Intercept) | -0.1235 | 0.6911 | -0.1787 |
| BMI | 0.0112 | 0.0079 | 1.4122 |
| WHR | 0.9381 | 0.7168 | 1.3087 |
| stab.glu | 0.0295 | 0.0010 | 29.9531 |
| chol | 0.0042 | 0.0012 | 3.5121 |
| hdl | -0.0048 | 0.0031 | -1.5429 |
| hypertension | -0.1434 | 0.1078 | -1.3304 |
| age | 0.0120 | 0.0034 | 3.5328 |
| time.ppn | 0.0005 | 0.0002 | 2.8592 |

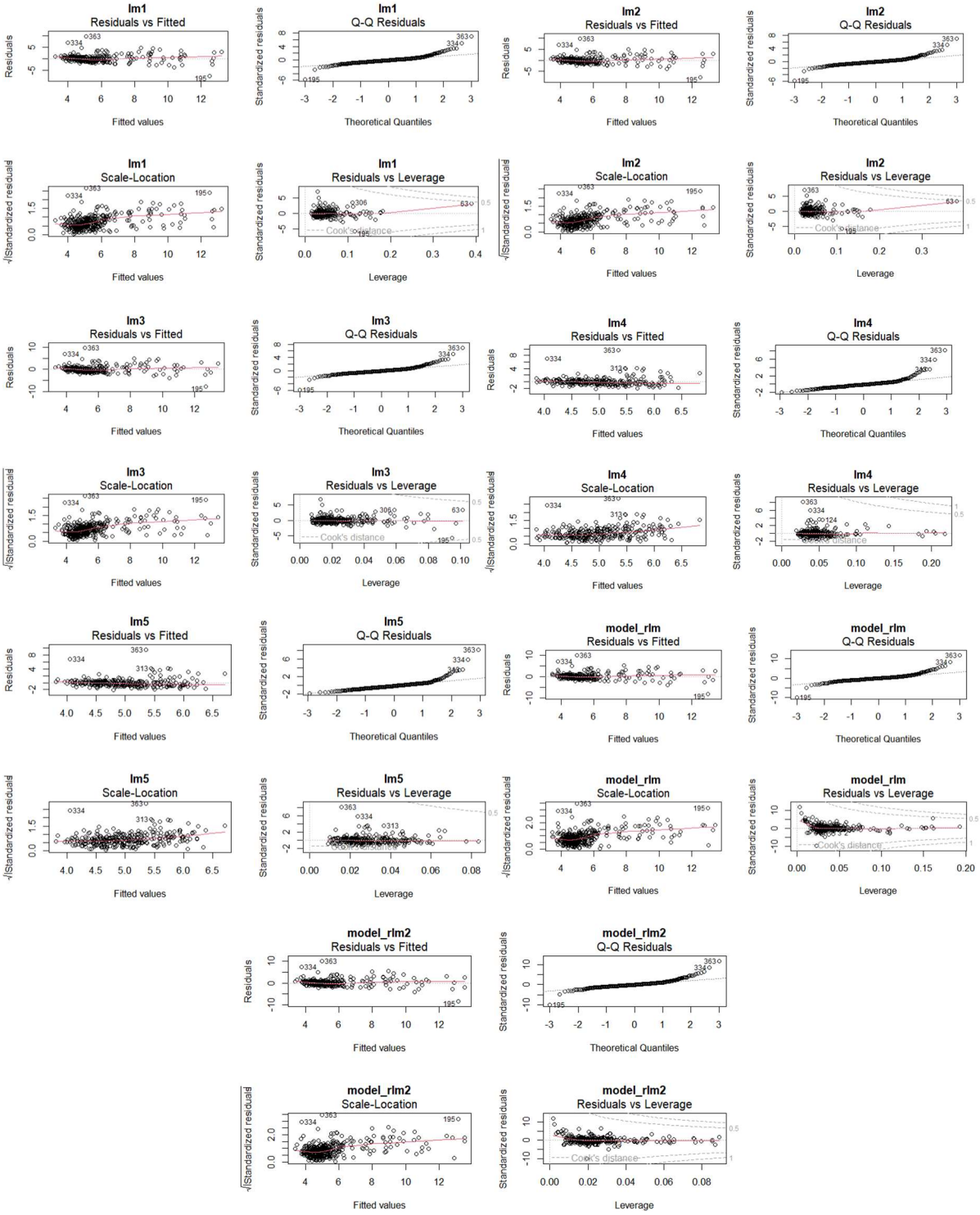
In this case, we can interpret the values of the t-statistic to see what coefficients are statistically significant. In particular, the threshold for the 5% significance level is around 2 in absolute value, therefore we obtain 4 statistically relevant coefficients in both models, which resembles certainly an improvement with respect to the non-robust models, even if not that huge. The relevant coefficients are similar to the ones of the previous models: we in fact have the level of glucose, the age, the postprandial time, but we also get an unexpected significant impact of the location in the first, and of cholesterol in the second.

Overall, the results found are in line with the expectations formulated before the estimation of the models, even if we could also have expected some stronger improvements.

## 5.4 Models Evaluation and Comparison

Once all the models have been computed, we proceeded with some diagnostic and evaluation of the performance to assess the validity of the models and to make a comparison between them. Initially we started plotting 4 kinds of different graphs:

- Residuals vs values fitted by the model to check if they distribute around zero
- QQ plot of the residuals to assess normality
- Scale-location to check homoscedasticity
- Residuals vs Leverage, to check the influence of different points




As we can see, all the models' residuals seem to follow an almost linear pattern, suggesting a linear, although not perfect, relation between variables. Moreover, the values of the residual are all very close to zero, except for some points that instead have a large error, suggesting the presence of other outliers. Besides these points, the assumption of linearity seems to be respected.

The QQ plot shows that the majority of the points of every model seem to follow a Gaussian Distribution, even if for many of them we notice some divergence along the left and right

tails. However, we can see that for model lm4 and lm5, the ones computed on the clean dataset, the left tail is much less heavy, suggesting that the outlier detection and deletion have somehow improved the assumption of normality, making it more reliable. In the end, the hypothesis of normality seems to be approximately respected, even though it is not perfect.

The Scale-Location plot resembles a way to check for the assumption of homoscedasticity: under homoscedasticity we should have a red line almost flat, and the residuals should be equally spread as the fitted value increases. In our case, we see that the line is almost never flat but has a positive slope, suggesting an increasing variance, and this unfortunately suggests a violation of the assumption for our models. In order to be sure of the results, we also decided to compute the Breusch-Pagan test, that is another way to validate the assumption. The null hypothesis of the test is of null variance. The results are shown below:

- lm1: 0.001476
- lm2: 0.0007365
- lm3: 5.495e-05
- lm4: 0.827
- lm5: 0.6113
- rlm1: 0.0007365
- rlm2: 5.495e-05

Fortunately, we see that at least two out of seven of the computed models achieved a p-value greater than 0.05, making then the assumption of homoscedasticity reliable. More specifically the two models are again the ones that achieved better results in the previous analysis: lm4 and lm5, the ones computed on the clean dataset.

Lastly, the residual vs leverage plot helps in discovering if there are some influential subjects that are driving the results of the regression. In the first three cases it is more evident that some data points, like point 63, are influencing the results a lot and have also a large residual. In the case of the regressions run on the cleaned dataset we can instead see that there are some points that achieve a high value of residuals (the same one that can be seen on the right tail of the QQ plot), but we have no influential points that achieve a high error. In the end with the robust approach, we get even more refined results, with most of the points concentrated around zero value.

During the process of developing each of the models, the Variance Inflation Factor (VIF) has been checked, to consider multicollinearity and select what variables should or should not be included in the specifications, exactly as done in the logistic section. Therefore, the choice of the variables used in the models discussed in the previous section was mainly based on this kind of analysis. The results obtained are the following:

```
                GVIF Df
chol        3.895473  1
stab.glu    1.209142  1
hdl         5.662383  1
ratio       7.037279  1
location    1.141019  1
age         1.783788  1
gender      2.678877  1
height      2.476398  1
weight      8.156910  1
frame       1.816321  3
systolic_bp 2.248883  1
diastolic_bp 1.864328 1
waist       5.615268  1
hip         6.952339  1
time.ppn    1.065688  1
```

```
                GVIF Df
chol        3.850234  1
stab.glu    1.192450  1
hdl         5.635367  1
ratio       6.994356  1
location    1.104337  1
age         1.388092  1
gender      1.434131  1
BMI         1.632606  1
frame       1.720163  3
WHR         1.402783  1
hypertension 1.208658 1
time.ppn    1.055442  1
```

The two models' specifications that use most of the variables in the dataset show many VIF's value that exceed the typical threshold of 5: for the first we have hdl, ratio, weight, waist, and hip, while for the second we have hdl and ratio. For this reason in the following model, we decided to maintain only numerical variables, excluding also ratio that was causing some problems. The results obtained in the third model have in fact improved significantly.

```
=== VIF: lm3 ===
      BMI          WHR     stab.glu hypertension         chol          hdl          age
 1.143774     1.150802     1.167301     1.189620     1.169864     1.184623     1.314066
 time.ppn
 1.012636
```

In this case, no multicollinearity is evidenced. For this reason, the best model specification in our study is for sure the one obtained by excluding categorical variables, that seem not to contribute to our goal, and the variable ratio, that was causing the majority of the problems in terms of multicollinearity.

Finally, we decided to compute some metrics to choose the best model among all the ones computed. The metrics included are the Akaike Information Criterion (AIC), the Bayesian Information Criterion (BIC), and the adjusted $R^2$. The first two indexes are used to evaluate the tradeoff between goodness of fitting and complexity of the models, while the adjusted $R^2$ is a measure used to understand how much of the total variability can be explained by the models, adjusted for the fact that they all have different number of covariates. (Note that adjusted $R^2$ is not available for the robust models because they deliberately avoid the squared-loss approach, meaning the standard decomposition of variance no longer applies).

```
   Modello       AIC      BIC    Adj_R2
1      lm1 1350.5852 1425.197 0.5854586
2      lm2 1346.1115 1408.942 0.5872365
3      lm3 1341.1500 1380.419 0.5863109
4      lm4  988.0720 1047.597 0.1585160
5      lm5  979.5078 1016.711 0.1662334
6  model_rlm 1353.4509 1416.282        NA
7 model_rlm2 1348.5117 1387.781        NA
```

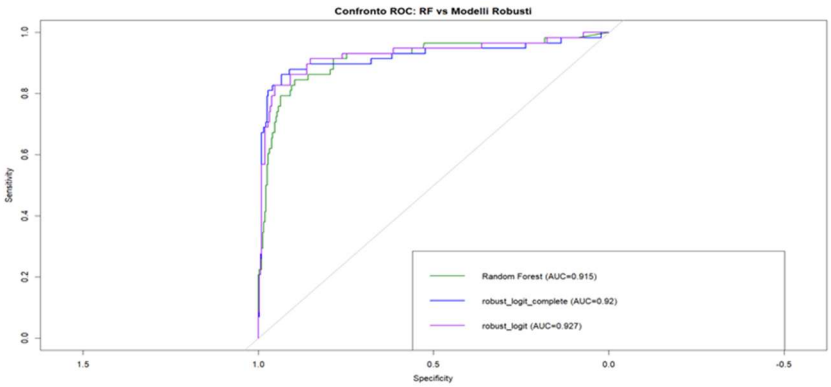
The best models in terms of Information Criteria are lm4 and lm5, that achieved very good results. However, we see a very consistent drop in the adjusted $R^2$, that is way

better for the other models, and this can be due to the outlier removal. In fact, deleting outliers removes also some variability in the data, that could maybe be something that the simplest models were able to explain easily. Yet, as seen in the diagnosis, the only two models that seemed to respect the assumption of the linear regression were lm4 and lm5, even if their $R^2$ is not that high, and even if the number of significant coefficients found by these models is lower than lm3. The best among the two is surely model lm5, because it does not suffer of any problem of multicollinearity, and because a model that does not respect the basic assumptions upon which it is build is not reliable, and any inference done on it can be meaningless, even if it seems to achieve good performance.

To conclude, in the case of linear models, we saw that a robust approach can lead to better results in terms of significance of the results and in terms of assumptions of the models. In this specific case, the best model was the standard one, but applied on a dataset cleaned by all the outliers found by MCD. This strategy worked better with respect to the robust linear regression. Therefore, at least for our specific case, using a robust method to find and delete anomalous individuals from the dataset helped achieve better outcomes.

## 5.5 Random Forest vs Robust Logistic models

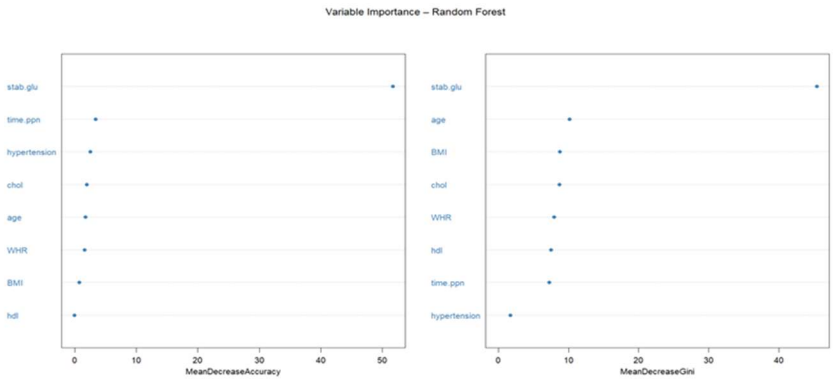
To find patterns that a linear model could miss, we used a **random forest** with 500 trees. This method is useful for seeing how the variables interact with each other.
To check how well the model separates diabetic from non-diabetic patients, we used the ROC curve and the AUC, area under the curve.




By comparing the random forest to the robust logistic models, we could see that the robust logistic model with only numerical variable is more precise than the random forest and the robust logistic model with also the categorical variables. This could mean that the relationship between the variables and diabetes is linear or that the dataset wasn't big enough to make random forest better than the other models.
Despite this, all the three models perform very well since they show an AUC above of 0.915, meaning that the models have at least a 91.5% chance of correctly identifying diabetics and

non-diabetics in the whole dataset.
In particular, looking at the confusion matrix, the random forest was good at identifying non-diabetics patients, with an error of just below 3%, while it has more problems in recognizing diabetics patients, with an error of 40%, meaning that it doesn't recognize 4 diabetics out of 10. This is due to the fact that we have way more healthy patients than diabetics ones.

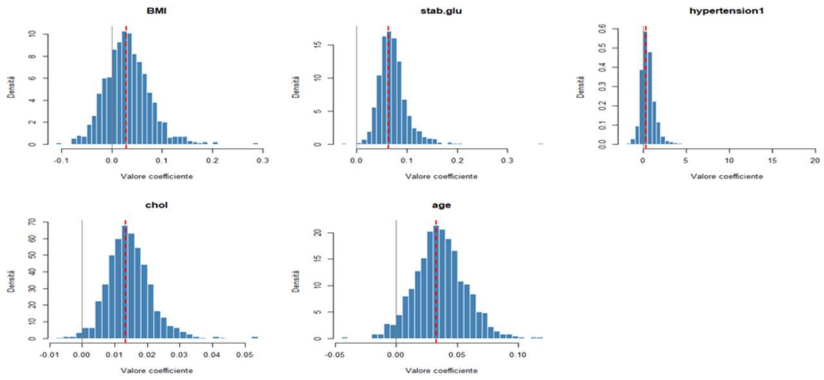
Predictor importance was assessed based on two criteria, the Mean Decrease Accuracy, which measures the decline in model accuracy when a variable's values are reorganized and the Mean Decrease Gini, which measures the reduction in the node impurity of a variable.




As we can see, the most important variable in the random forest is the stabilized glucose. This means that, in terms of accuracy, when we shuffle the values of stabilized glucose, the model's predictive power drops significantly more than for any other variable; in terms of Gini impurity, stabilized glucose is the most effective variable for splitting the data into clean "diabetic" vs "non-diabetic" groups at the tree nodes.
The other measurements are still important but way less than stabilized glucose. This highlights that while, for example, body composition, age and hypertension are risk factors, the direct measurement of blood is the best way to assess a patient status.

## 5.6 Bootstrap Method

We ran a bootstrap with 1,000 replications on the best coefficients we found to be sure that the results aren't random.
First, we calculated the confidence intervals for the variables. If the interval does not include zero, we can be confident the variable has a real effect.
Then we created histograms of the bootstrap results. The red line shows our original estimate, and the histograms show how much that estimate changes across different samples. This helps identify if any specific variable is unstable or biased.

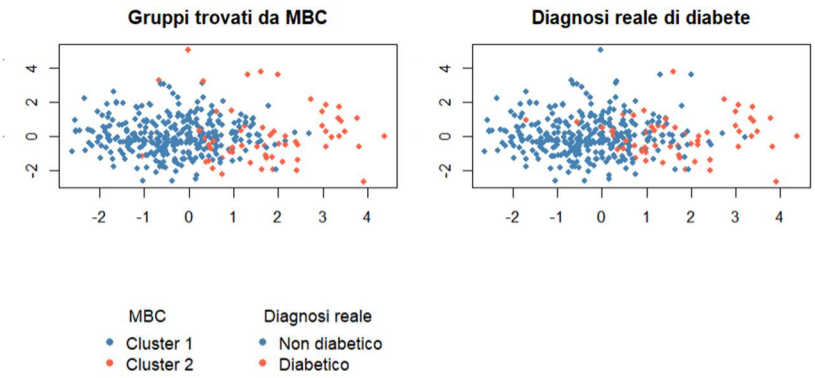



In this graph we can see that apart from cholesterol and stabilized glucose, all the variables are not significant since they have very high standard errors that bring the intervals to include the zero.
Hypertension must be treated differently. The number of patients that have hypertension and diabetes is more than double the number of patients with diabetes and no hypertension but, since the absolute number of people with hypertension is small, when we resample data the group changes too much, making the results unstable.
This is possible due to the fact that some variables have a too small pool of patients to make bootstrap effective.

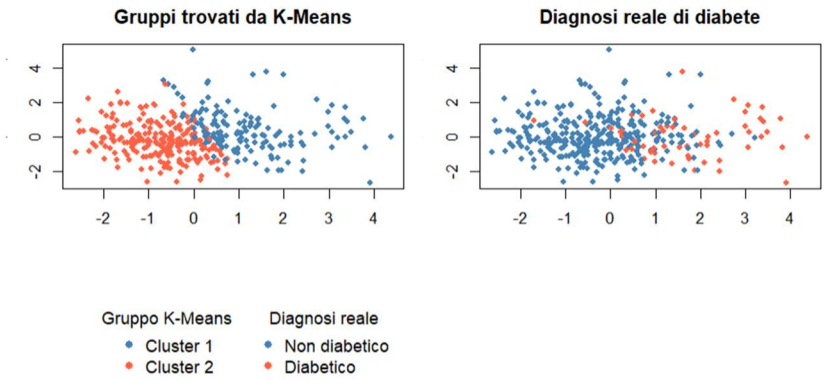
## 6. Clustering analysis

We also wanted to see if the data naturally forms clusters that match the actual diabetes diagnoses, and we did so by doing three different clustering methods:

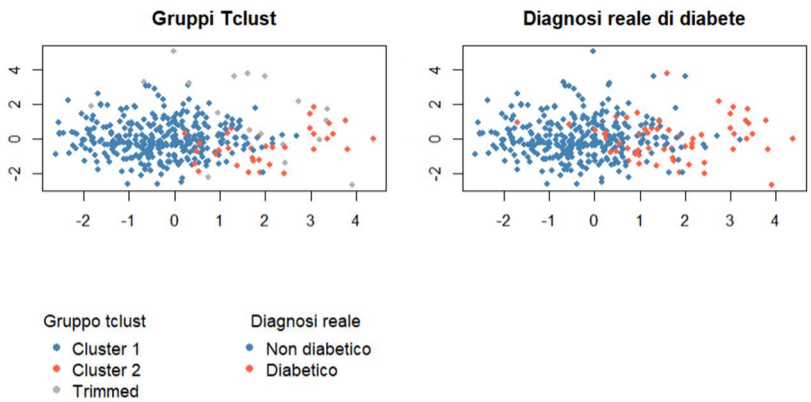
1. Model Based Clustering, which uses statistical distributions to find groups. BIC scores were used to find the best model, which is a VVE model (Variable volume, Variable shape, Equal orientation) with a BIC of -5884.




2. K-means, where we used elbow plot and silhouette analysis to confirm that choosing two clusters was the best option.




3. Robust clustering, which is similar to k-means but ignores the 5% of the most extreme outliers.




After this we computed ARI, the Adjusted Rand Index for all the clustering methods, which is a metric that measures the similarity between the cluster that "naturally" formed from the data and the actual clusters.

```
Riepilogo ARI
Mclust   (K=2)            : ARI = 0.5452
K-Means  (K=2)             : ARI = 0.1956
tclust   (K=2, a=0.05)     : ARI = 0.5904
```

As shown in the summary, robust clustering is the best model, achieving an ARI slightly above 0.59. Ignoring the most extreme 5% of the outliers spiked model's ability to identify diabetics and non-diabetics.
This is a good result considering that it's significantly higher than zero and we are working on clinical data, so we can say that robust clustering provides good evidence for the validity of our predictive variables.

## 7. Conclusions

We were able to demonstrate that diabetes is not only influenced by a single factor, but by many variables related to biochemical and anthropometric markers. Going more in depth, stabilized glucose is the primary driver of the patient classification, while HDL and total cholesterol, despite being less crucial, are still very important in the analysis.
The anthropometric markers, while being correlated with diabetes, have a weaker correlation than the other variables, but it's still important to address also to these variables during the analysis.
Looking at the robust logistic regression on the non-cleaned dataset, we can confirm the impact of the variables cited before, with the addition of three other variables: gender, hypertension and the post prandial times when labs were drawn.
The post prandial times when labs were drawn are important since we must know if the patient had his labs drawn exactly after 2 hours after starting the meal, so that blood sugar management is correctly measured.
Hypertension is not as significant as the other variables, but it's still worth mentioning since people with hypertension have a probability of having diabetes 5.5 times higher than those without hypertension.
In this sample, the coefficient for males is negative, meaning that, ceteris paribus, females have a chance of having diabetes 7.7 times higher than males. Digging in the dataset, this phenomenon is clarified by the average ages of the genders: healthy women are younger than healthy man (43.5 vs 46.8 years), while diabetic women are older than their counterpart (59 vs 57.8 years). The 15.5-year gap could suggest that a possible reason for this difference could be due to post-menopausal metabolic shifts, that the model can capture only in the gender variable.
For linear regression models, the best model is lm5, so the one done on the dataset without outliers, removed with the MCD, and without categorical variables and ratio.
Its R2 is not that high, but it's the only linear model that simultaneously doesn't show multicollinearity problems and that respects the assumption of linear regression, without which a model cannot be considered reliable and, therefore, meaningful.
In the clustering part, we saw that, while k-means was inadequate, model-based clustering and the robust clustering were good in differentiating diabetics and non-diabetics patients, with robust clustering being the most precise cluster model. This was an expected result since, in clinical studies, removing atypical patients is fundamental to obtain a good understanding of the underlying disease patterns.